

Audio/Video Guidelines for Assemblies and Conventions Addendum

August 2025

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CHAPTER 1 INTRODUCTION

1. This addendum is designed to be used in conjunction with *Audio/Video Guidelines for Assemblies and Conventions* (CO-160). It contains direction that applies specifically to the New Zealand region of the Australasia branch territory, such as how to obtain assistance from the Local Broadcasting Department Help Desk and instructions on how to set up and operate specific audio/video equipment.

CHAPTER 2 TECHNICAL SUPPORT

1. To obtain assistance, please contact the Local Broadcasting Department Help Desk. See the contact information and hours of operation listed below. For events held at Assembly Halls, work with the technical contact at the facility before contacting the help desk.

(1) **Telephone Number:** 00 61 2 8203 9304

(2) **Email:** LBD.au@bethel.jw.org

(3) **Hours of Operation:** Monday-Friday 8:00 a.m. to 5:00 p.m. AEST

(4) **Note:** If no one can be contacted using the above number, an alternate number that can be used after hours during the convention season is: 00 61 2 8203 9305

2. During the convention season, the help desk provides additional support during the hours of operation listed below:

(1) **Additional Hours of Operation:** Friday-Sunday 7 a.m. to 5 p.m. AEST

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Audio Headend

1. The standard Audio Headend provided to locations in New Zealand by the Australasia Local Broadcasting Department includes the following equipment:

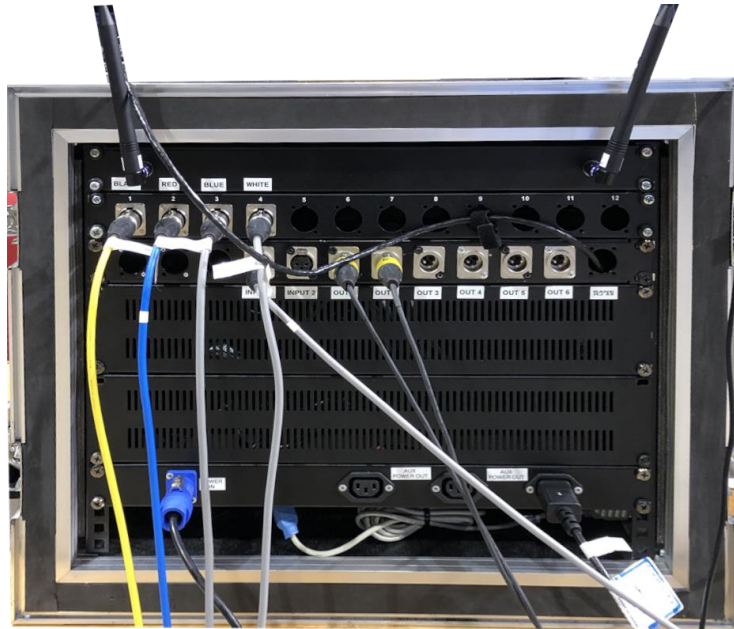


- (1) 1 x Furman Power Conditioner to power the systems in the Headend.
- (2) 1 x Beyerdynamic Antenna Amplifier system.
- (3) 1 x Beyerdynamic NE 914 Audio Receiver.
- (4) 1 x Rane Audio Digital Signal Processor.
- (5) Nine Microphones:
 - 4 x Beyerdynamic Wireless Condenser Microphones.
 - 2 x Beyerdynamic XLR Hypercardioid Microphones (M88 TG).
 - 3 x Beyerdynamic XLR Cardioid Microphones (TG V56C).

2. The headend is powered by the single Furman Power Conditioner. The headend can be turned on and off using the one switch on the power conditioner at the top of the rack. Each device does not need to be turned on and off individually.

3. How to set up the Headend:

- (1) On the top row at the back of the headend unit, screw in the two Mipro AT-90W Antenna Receivers.
- (2) On the second row connect the four wireless microphone outputs to the audio mixer.
- (3) The third row manages connections for the audio processing. Each input and output are configured for a purpose.



- (4) Input 1 drives outputs 1-5:
 - Output 1 and 2 are configured to provide two Front of House signals.
 - Output 3 and 4 are configured for In House speakers.
 - Output 5 is configured with a delay for the FM system.
- (5) Input 2 drives output 6:
 - This channel is for the stage monitors (foldback).
 - The separate input allows the audio to the stage monitors to be equalized separately to reduce feedback. Level is also controlled separately by a Group Output.

Audio Receivers

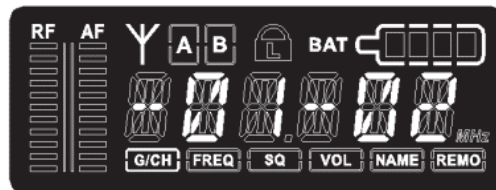
4. **Beyerdynamic NE 914 Audio Receiver.** This is an audio receiver for the four provided Beyerdynamic wireless microphones. The antennae are run through the amplifier to ensure a strong signal to this device.



- (1) The left knob (1) sets the volume for the headphones, and can be clicked in to configure which headphones are being monitored (by the headphone jack below the knob).
- (2) To select a channel for configuration, hold the menu button in (2), until the green light on the currently selected channel (3) starts flashing. Turn the menu control to select

the receiving channel. Press the button in once more and that channel will be selected.

- (3) To edit a setting on the channel, rotate the knob until the relevant setting is highlighted. In the picture below the G/CH channel selector setting is highlighted.



- Press the knob in to enter configuration mode for that setting.
 - Rotate the knob until the settings are correct.
 - Press the knob in again to confirm the changes.
- (4) The Scan button below the channel display can also be used to select a channel. Press the Scan button once and the receiver will adjust to an interference-free channel automatically within the selected group. Press the menu control to confirm the selected channel.
- (5) When the channel/frequency settings are adjusted, the corresponding microphone will need to be reconnected.
- To connect a microphone, press the ACT button below the channel display on the receiver, then hold the microphone within 20cm of the receiver unit.

5. **Beyerdynamic ZAS 900 Antenna Amplifier.** This amplifies the signal from the Antennae for the wireless microphones.

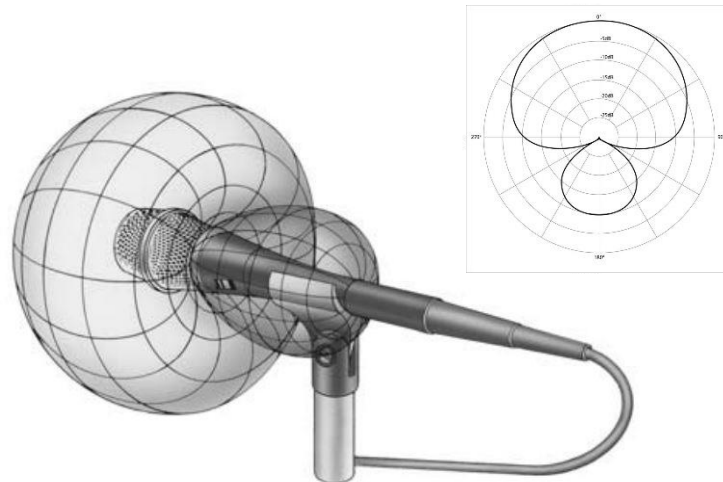
Digital Signal Processors

6. **Rane HAL3s Digital Signal Processor.** This is designed to take the output of the audio mixer and apply several audio processes to improve the quality of the audio. This is fully configured by the branch. The connections involved are detailed in the Audio Headend section.

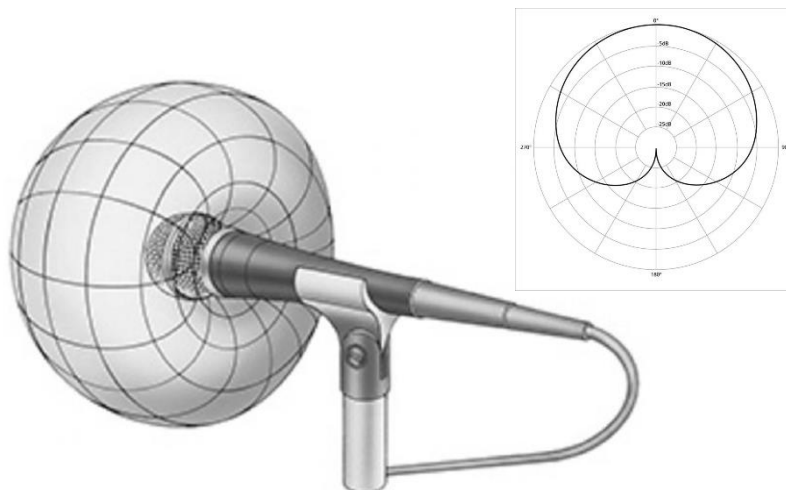
Microphones

7. **Polar Patterns.** A polar pattern is the sensitivity the microphone has to sound coming from each direction. The microphones provided use the Super Cardioid polar pattern, the Hyper Cardioid polar pattern or the Cardioid polar pattern. In all of these cases, the top of the microphone should be directed towards the person speaking, and at a distance of 10 to 15 cm for the best audio quality.

- (1) **Super Cardioid Polar Pattern.** This style of microphone is very good at rejecting noise from the side of the microphone.



- (2) **Hyper Cardioid Polar Pattern.** This is essentially the same as a super cardioid polar pattern, but with a narrower pickup range.
- (3) **Cardioid Polar Pattern.** This style of microphone is very good at rejecting sound from directly behind the microphone.



8. **Beyerdynamic S 910 M-81.** This is a wireless handheld Condenser Microphone, with a cardioid polar pattern. These are colour coded to make identification easier for the audio mixer brother. Spare batteries are supplied in the rack drawer. For connection instructions see the audio receivers' section.



9. **Beyerdynamic M88 TG.** This handheld microphone has an XLR Male connection. It is a dynamic moving-coil microphone, which gives it a Hyper Cardioid polar pattern. It must be connected to the mixer by XLR cable.



10. **Beyerdynamic TG V56C.** This has an XLR Male connection. This is a Cardioid Condenser Handheld Microphone.



Stage Boxes

11. A stage box is a remote interface device used to connect equipment to a mixing console. It provides an on-stage central hub for connecting microphones, instruments, and powered speakers or amplifiers to a multicore cable snake or a digital cable. This allows the sound desk to be further from the stage. Instead of having to route several cables across a long distance, only one cable needs to be routed. Analogue stage boxes don't convert the signal, just transport it through a multi-core cable. Digital stage boxes convert the signal from analogue to digital, and then transport the digital signal, usually via ethernet or fibre optic cables.

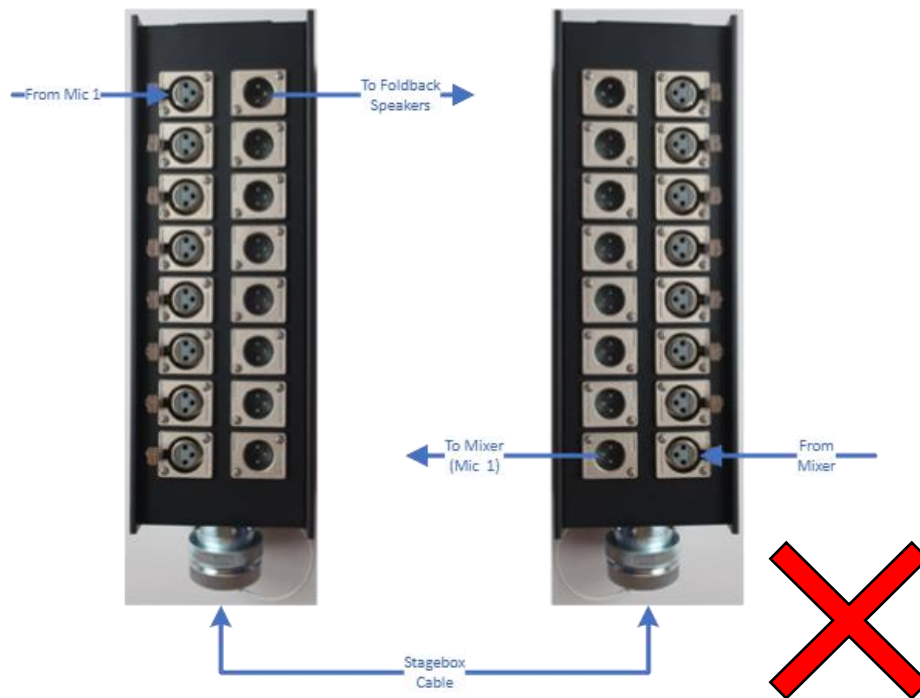
12. **Canare Stage Box pair.** The Canare Stage Box Pair is analogue. It uses a multi-core cable to transmit the signals without conversion.

- (1) This system is made up of a matched pair of Stage Boxes and a multicore cable between them. There are two 50m cables provided with the kit that can be connected to create a 100m connection.
- (2) **Canare 8J12N1 8 Channel XLR Female Stage Box.** This is the Female Stage Box component of the kit. (Image Left).
- (3) **Canare 8J12N2 8 Channel XLR Male Stage Box.** This is the Male Stage Box component of the kit. (Image Right).

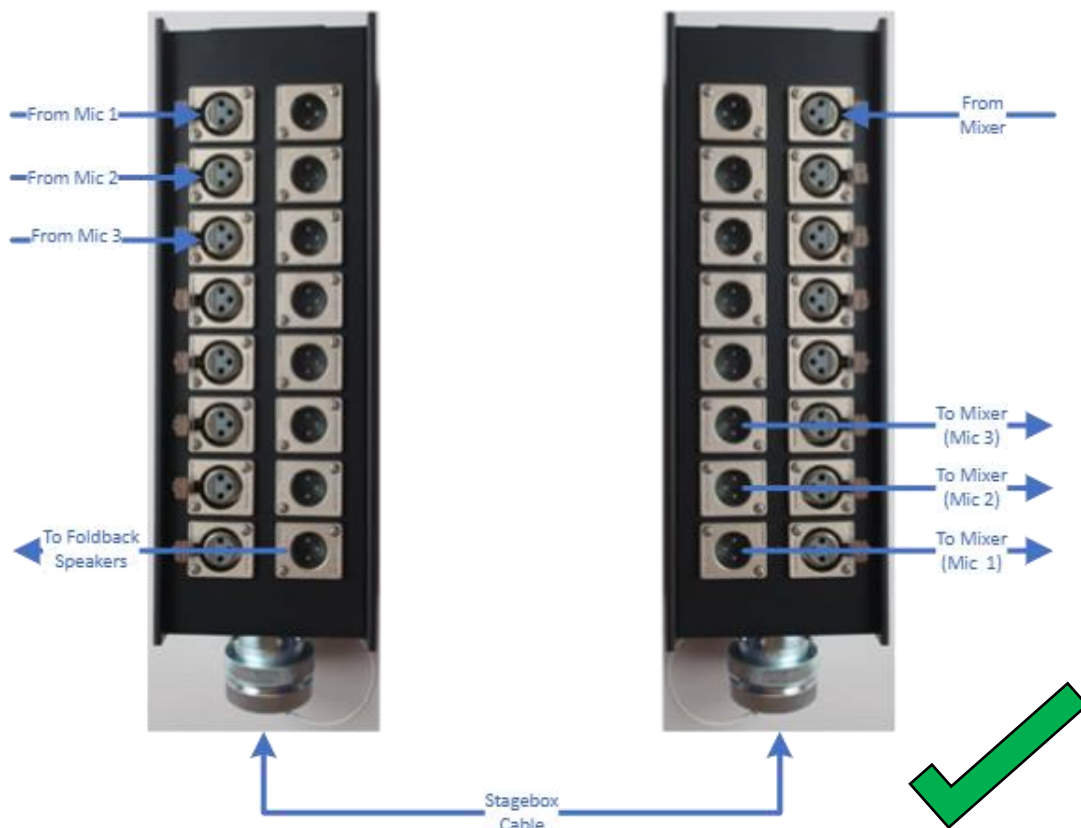


- (4) Each of the eight channels has parallel connections to the Male and Female XLR access points on both sides.

- (5) **Note:** The corresponding male and female XLR ports on one stage box for a given channel cannot be used at the same time, as they connect to the same underlying cable/channel. The below configuration is an example of a **bad** stage box setup. The input and output signals will interfere with each other and possibly cause damage to the equipment.



- (6) Valid Stage Box connection diagram. This is just an example, always follow the AV designs provided.



Speakers

13. **Mipro MA-808 Public Address portable speaker system.** This setup is used for the School for Kingdom Evangelizers (SKE), but can also be used as a PA system for emergencies. These can be run off mains power, but also come with a battery and so can be used portably when needed. These have Balanced, unbalanced, mic and line-in inputs. They also have a compartment containing a wired XLR microphone that can be used directly with the speakers.



Stands

14. **Beyerdynamic GST500 Microphone Floor Stand with Boom Arm.** This stand features a telescopic boom arm.

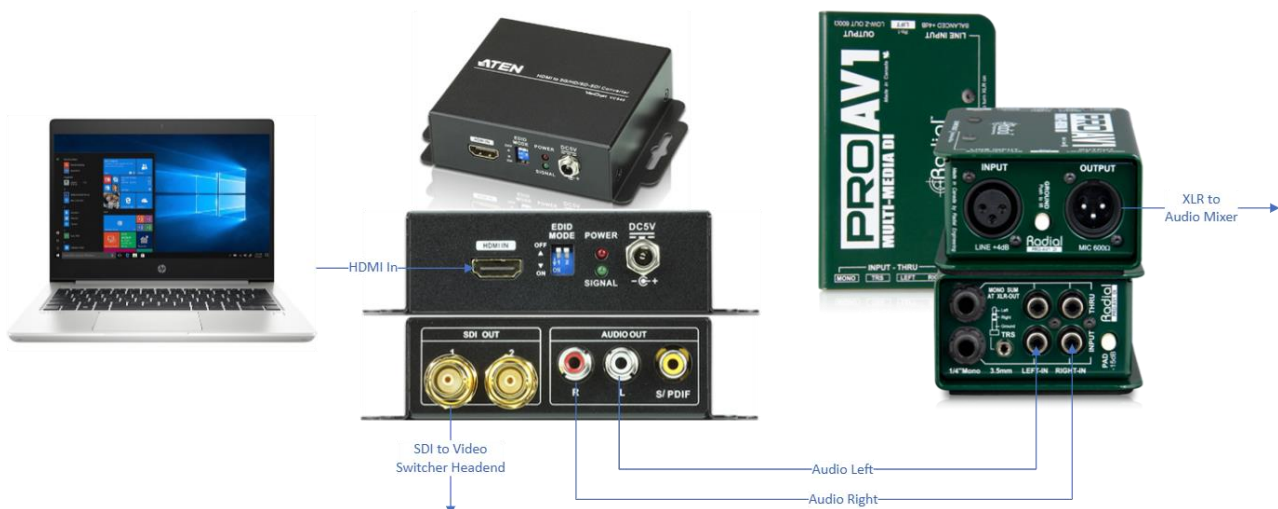
Audio Converters

15. **Australian Monitor - BUZZ OFF Hum Eliminator.** The BUZZ OFF is ideal for getting rid of hum and extraneous noise from long cable runs. The device contains two channels. Each channel can be used to process a balanced XLR signal or an unbalanced RCA signal, and will output to the corresponding balanced or unbalanced outputs.



16. Setting up the ATEN HDMI Converter and the ProAV1 DI Box. This is the normal setup used for extracting an audio and video signal from the media devices. The Converters used are described below.

- (1) Connect the laptop to the ATEN Converter Via HDMI.
- (2) Connect the Left and Right RCA AUDIO Out signals from the ATEN Converter, into the LEFT-IN and RIGHT-IN INPUT connectors on the ProAV1.
- (3) Connect the XLR OUTPUT signal from the ProAV1 box to the Audio Mixer.
- (4) Connect the SDI Output 1 from the ATEN Converter to the Video Switcher.



17. ATEN HDMI to 3G/HD/SD-SDI Converter. This is used to take the HDMI signal from the laptops, and convert them to an SDI video only signal and to left and right RCA Audio Signals.



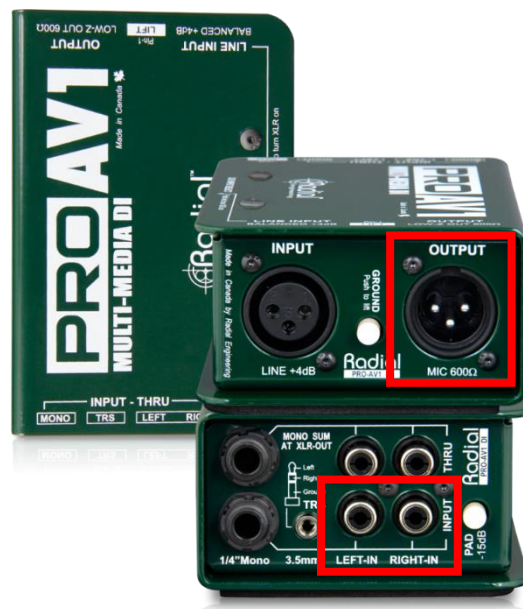
- (1) The left and right RCA signals will be converted to a balanced single channel XLR signal through one of the ProAV1 boxes.

- (2) Ensure the EDID MODE is set to ON for both channels. This enables a protocol that allows the display device to communicate its resolution requirements to the computer. This means that a signal failure or a blurry signal due to resolution mismatch is less likely to occur.



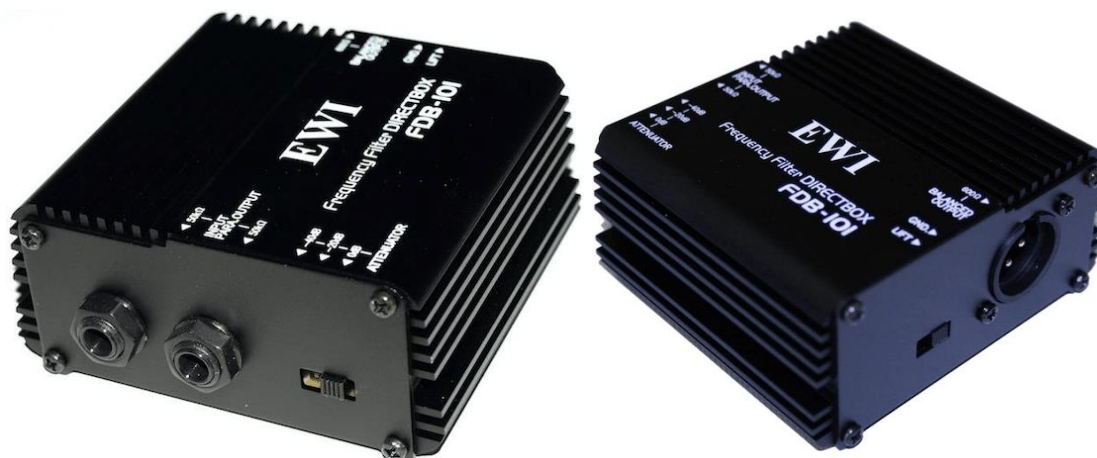
- (3) If there is no signal on the video switcher:
- Check these switches are toggled.
 - Check the signal coming from the laptops is at a resolution of 720p and a refresh rate of 50i or 50.
 - If the resolution or frame rate cannot be corrected, as an emergency fix, the HDMI signal can be run through a Decimator MD-HX HDMI/SDI Cross Converter to force the desired resolution settings (the configuration would be Laptop – Decimator – ATEN).
- (4) See the diagram under Chapter 3, Audio Converters, 'Setting up the ATEN HDMI Converter and the ProAV1 DI Box'.

18. Radial ProAV1 DI Box. The ProAV1 provides the ability to connect almost any audio source to a PA system, while automatically converting stereo sources to mono to preserve channels and simplify the audio setup. For conventions, these are used to convert left and right RCA Signals from the ATEN converters, into an XLR signal for use by the Mixer.



- (1) Plug the left and right RCA Cables into the LEFT-IN and RIGHT-IN INPUT channels respectively.
- (2) The signal for the Audio Mixer will come out the 'OUTPUT' XLR male port.

19. **EWI FDB-101 Direct Input Audio Converter.** This takes unbalanced ¼" Jack input, and outputs a balanced XLR signal. This converter would be used to convert an audio source into one compatible with the mixer. It can also be used to convert an audio signal into balanced to make it more suitable for transmission over a distance.



- (1) The switch on the left of the device can be used to set a level of attenuation on the signal. Leave this in the 0DB position (slid to the right).
- (2) The switch on the right side of the device determines if the signal will be grounded or not. Setting the switch to 'LIFT' will disable the grounding functionality. In general, keep the device set to GND (ground).

20. **Klark Teknik DI 10P Passive DI Box.** This takes unbalanced RCA input, and outputs a balanced XLR signal. This converter would be used to convert an audio source into one compatible with the mixer.



- (1) The knobs on the input side of the device can be used to set a level of attenuation on the signal. Leave these set clockwise (at 3PM) to give the full signal to the mixer.
- (2) The switch on the output side of the device determines if the signal will be grounded or not. Setting the switch to 'LIFT' will disable the grounding functionality. In general, keep the device set to EARTH (grounded).

Communications Equipment

21. **HME DX100 Wireless Intercom system.** This system allows the AV Team and the Stage brother to communicate effectively. The official manual 'HME DX100 Operating Instructions' can be found online. The system involves the following pieces of equipment:

- (1) **HME BS100 Base Station.** This is the hub that receives and sends signals from the belt packs. If there are range or signal quality issues, check that the antennae are connected properly. The antennae should be either one vertical and one horizontal, or both pointing vertically upwards in order to ensure a complete coverage. This system contains a battery compartment, but for convention use it should always be plugged in to power. There are two sets of antennae available for use:
 - There is a small black set which should be sufficient for smaller venues.
 - There is a set of long white antennae which can be used for larger venues.
- (2) **HME HS14 Headset Microphone.** This headset **does not** work standalone – it must be connected to one of the BP200 Belt Packs.
- (3) **HME BP200 Belt Pack.** This controls the headset.
- (4) **HME Battery Charger.** The HME charger has six storage slots (left) and four charging slots (right). When plugging in batteries to be charged, a Red light indicates the battery is actively being charged, and a green light indicates the battery is fully charged. A flashing yellow light with an inserted battery indicates an overheated battery, if this occurs move the system into a cooler area. A steady yellow light indicates the charging failed, if this occurs follow the troubleshooting instructions on the side of the battery charger.



22. Operating the HME System:



- (1) The Volume up (1) and Volume down (2) buttons are used to control the headset volume.
- (2) Pressing Volume up or down while holding down ISO (4) will alter the microphone gain to accommodate louder and softer speakers.
- (3) The headsets can be set to either Push-To-Talk (PTT) or Hands-Free (HF) mode operations. Push-To-Talk mode will require the operator to hold down either the ISO (4) button or the IC (5) button to talk. Hands Free mode will have the mic broadcast whenever the operator talks, not requiring any buttons.
- (4) Toggling between PTT and HF modes. A quick press of either the IC or the ISO buttons will make the device toggle between the two modes.
- (5) Note: All belt packs are configured so that the ISO and IC buttons function the same.

FM Transmitters

23. **Ramsey FM35 Audio Transmitter.** This is the FM Transmitter used in the New Zealand kit.

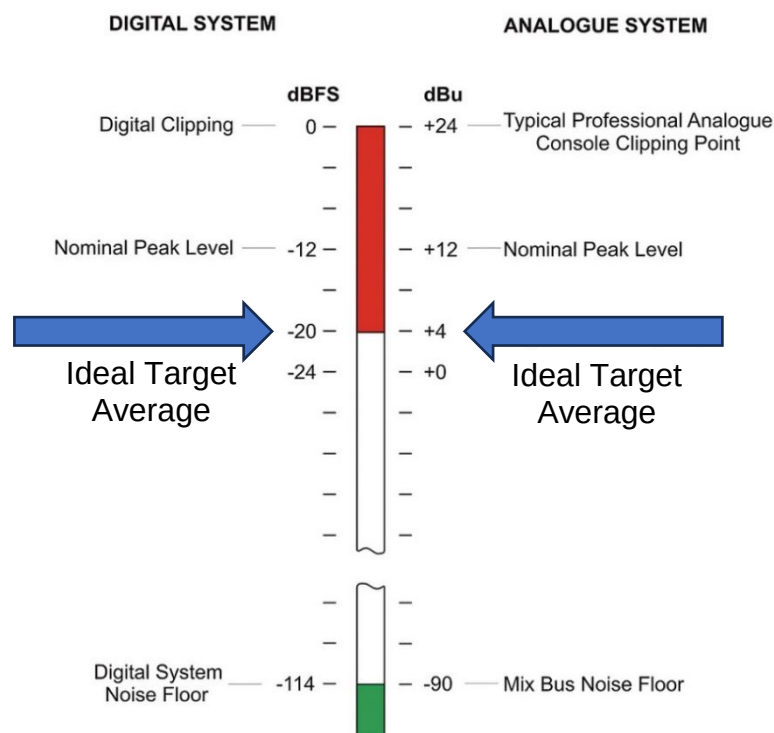


- (1) Always have the antennae plugged in before turning on the device.
- (2) The Setup button cycles through the various screens, and the Up/Down buttons adjust the value on the given screen.
- (3) Press setup until the frequency window is selected, then set the frequency to that specified in the FM license.
- (4) The next window toggles between mono and stereo audio.

Audio Mixers

24. Adjusting gain using solo and the LED meter. When setting up a mixer, it is best to adjust the gain for each input to ensure a good signal level with the channel fader at unity. The audio level is displayed on the mixer's LED meter, which differs for digital and analogue mixers. The goal when adjusting the input gain is to have the audio level averaging around -20 for a digital mixer and around +4 for an analogue mixer. To perform this adjustment:

- (1) Have the output faders lowered.
- (2) Press SOLO on the channel being adjusted.
- (3) Start with the microphone inputs. Have someone with a loud voice read a passage to generate audio into the microphone for that channel.
 - If the mixer has a pre-fader level (PFL) solo mode, the fader can be left down. If PFL can't be enabled on a given mixer, raise the channel's fader to unity (0).
- (4) Adjust the input gain knob clockwise until the meter reaches around the target level.
 - Avoid turning the input to gain to maximum. This can cause noise or hiss.
 - Even with the gain almost maxed, the audio level might still not be at the desired target level. If this happens, take note of the level the microphone can reasonably achieve, and use this level as the basis for adjusting the other gain knobs.



- (5) Adjust the input gain for the media inputs. With pink noise or prelude music playing, and the computer volume set at 80%, adjust the input gain knob clockwise until the meter reaches around the same target level as the microphones.
- (6) The topmost light in the LED meter indicates channel clipping (it might be labelled 'clip' or display a number to indicate the clipping level). If this LED ever illuminates during testing, reduce the input gain for the active channel.

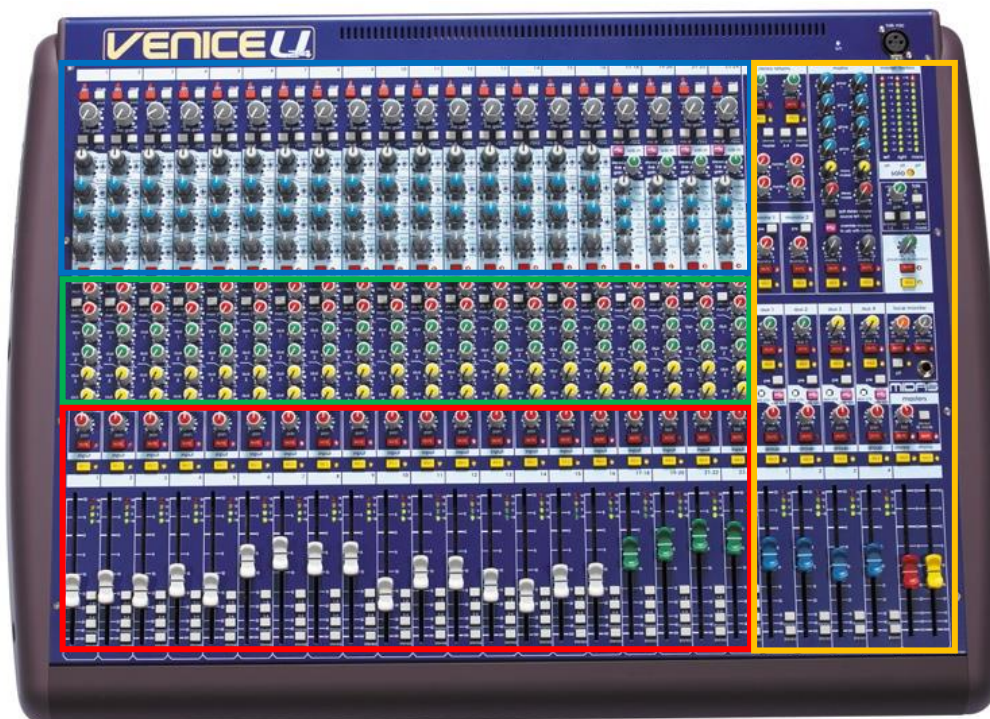
25. Adjusting Mixer Output Levels. Different pieces of audio equipment have different input and output level requirements. Depending on what equipment is next in the audio chain, the input or output levels might need adjustment. If the next device expects a signal level significantly lower than the mixer is outputting, it might clip or distort the sound. If the next device is expecting a signal level significantly higher, to compensate, it may need to add gain and perhaps introduce hiss or noise into the system. Work through each device in the signal chain to ensure the signal is neither too high nor too low for the next device. One way to achieve this is:

- (1) Connect each audio device in the chain.
 - Mixer to Processors, Processors to Amplifiers, Amplifiers to speakers.
 - Some of these devices may be integrated or pre-managed in a headend or rack, not needing to be connected during setup.
- (2) Make sure all amplifiers are turned down.
- (3) Bypass all processing (Graphic EQ, DSP). Note: Once these are engaged, gain may need to be raised slightly to compensate, if possible.
- (4) Have someone read a passage in a loud voice, or play prelude music.
- (5) Raise the fader for that input channel to Unity (0).
- (6) Slowly raise the master fader to Unity (0).
 - At some venues inhouse amplification is used, and some of these have the amplification set to a predefined level they don't want adjusted. In these cases, the output faders will need to be adjusted to whatever level suits the inhouse system.
- (7) Check that the signal lights on the equipment in line after the mixer are also showing good levels.
 - These should match the levels shown on the mixer meter, and not be clipping.
 - The fader might need adjustment at this point to ensure the signal going out is at the level needed by the next device.
- (8) Slowly raise the amplifier or speaker level.
- (9) Listen for excessive noise, hiss, or any distortion.
- (10) If there are any issues:
 - Check each device again to see if any are clipping, or getting too low a signal.
 - Check if any of the gain knobs are maxed out or close to being maxed out. (this can cause issues, just dial it back slightly).

26. Midas Venice U24 Audio Mixer.



27. **Midas Venice U24 Audio Mixer.** This is one of the Audio Mixers used to manage all the needed audio inputs for conventions in New Zealand. It will be used to manage audio from the XLR Cabled Microphones, from the Wireless Microphone Receiver system, and from the Laptops. Ensure each device is connected to the correct input channel so that the labels on the faders match the layout of the equipment. This guide divides the control surface of the mixer into four sections. Three input sections: **Fader Area**, **Auxiliary Send Controls** and **Audio Equalising Controls**. It also covers the **Output Section**. And the Rear output panel. For more information, the full manual for the Midas Venice U24 Audio Mixer is available online.



28. Input Section – Fader Area.

- (1) **The Fader.** This controls the volume of the input.
- (2) **The Mas, 1-2, 3-4 mono and stereo buttons.** These buttons can be used to make an input be sent to only a subset of the output channels. For example, some inputs may not need to be sent back to the stage monitors. In normal operation these three buttons will all be pressed in.
- (3) **The Solo button.** This makes the LED meter on the top right display the levels for this input only. It can be useful when adjusting the volume for an input.
- (4) **The Mute button.** This mutes the input.
- (5) **The pan control.** This controls the degree to which the input feeds the left channel, the right channel, or both. For inputs, leave this at 0 (feeding both left and right channels).



29. **Auxiliary Send Controls.** There are six auxiliary send outputs. There are mon 1 and 2 (1), aux 1 and 2 (2), and aux 3 and 4 (3). On each input there are knobs that control if the input will be sent to the given auxiliary channel.

- (1) **Mon channels.** These can be configured for pre or post fader operation per input. Press the button in to enable this behaviour. These output to the monitor 1 and monitor 2 outputs on the rear of the audio mixer.
- (2) **Aux channels.** These can be configured for pre or post fader operation (see the output section for more information).

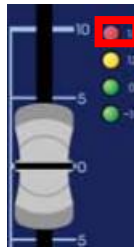


30. **Audio Equalising Controls.** These are useful for creating natural sounding voices, ensuring high quality audio from media, and reducing problems with feedback and pops or sibilance.

- (1) **EQ on.** The equalisation settings will only activate if the red eq button is pressed in.
- (2) **EQ knobs.** Each ring has two knobs. The outer knob selects a frequency, and the inner adjusts the audio/decibel level around the selected frequency. With each knob pointed up at 12 o'clock, there will be no effect on the sound. There are four targetable frequencies per input on the mixer. Avoid being overly aggressive with the equalisation settings, as this can result in the audio sounding unnatural.
- (3) **Mic phase switch.** This causes a 180-degree phase change in the input signal, reversing the polarity. This is commonly needed where two microphones are used facing each other. Ordinarily the two microphones would be out of phase causing cancellation when the console sums the two signals into the output. Reversing the phase of one signal causes the microphones to be out of phase with each other and no have no cancellation.
- (4) **80Hz.** Enabling this attenuates any audio signal below the 80Hz frequency.
- (5) **Gain.** This adjusts the input amplifier gain before the signal is adjusted by the fader.
- (6) **+48v / phantom power button.** **Never press this button when the fader is up.** This button activates phantom power to the XLR connector. See the microphones section for information about which microphones need phantom power.



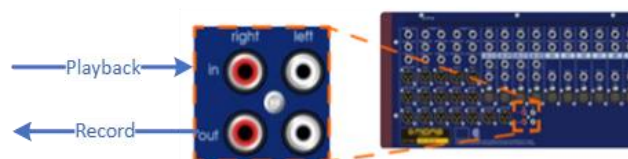
- (7) **20dB pad switch.** This provides 20dB attenuation on the input signal, allowing for high level signals to be used without overloading the input amp. Press this if there is frequent flashing on the red LED next to the fader.



31. Output Section – Stereo, monitor, matrix and master meters.



- (1) **LED meter section.** This section aids in setting the correct volume levels. When a solo button is engaged the two left columns of LEDs will display the signal level of the selected solo source. The system displays the after-fader levels for the left and right channels, and displays the pre-fader levels in the mono channel.
- (2) **Talkback section.** Press on to enable talkback. The Gain knob sets the attenuation on the talkback signal. The bottom three buttons determine which channels the talkback signal gets sent to.
- (3) **Playback.** Connect an analogue device here to play back recorded audio material via the console.



Playback/record sockets on the rear panel

- (4) **Matrix inputs.** These two columns determine how much of each output gets sent to the two matrix outputs.

- (5) **Matrix outputs.** These control the output from the matrix setup.
- (6) **Stereo Returns.** This section controls the Stereo returns. The knob sets the output levels of the stereo returns and has mute/solo operations. Below that are buttons to set which busses the returns get sent to.
- (7) **Monitor section.** This determines how much of each stereo return signal gets sent to the two monitor channels.
- (8) **Monitor Output section.** This has gain, mute and solo controls for the two monitor channels. The pre button, if pressed, will make the pre-fader signals get sent to the monitor channel.

32. Aux/Group output section.



- (1) **Aux controls.** These control the output volume of the aux channels, the mute and solo functions, and the pre or post fader operation.
- (2) **Aux c/o button.** This swaps the output paths of the aux and group channels. Specifically, pressing the button in will connect the aux bus to the group insert, group mute, group meter, group solo, group fader, group pan and finally the group output XLR. The group bus will then connect to the aux output pot and to the aux output XLR. This feature is meant for those that want additional controls on the aux channels instead of the group channels. This button should not be active during convention operation. A sharp object like a paper clip or a pen will be needed to toggle this button.
- (3) **Pan, mute and solo.** This section handles pan, mute and solo for the group channels. Pan sets what portion of the signal goes out the left and right channels. Making the pan knob point up to control will make the signal go out to both channels evenly.
- (4) **Headphone section.** Headphones can be connected here to monitor the output of the system. Pfl sets if the signal comes out is in pre-fader level mode.
- (5) **Master outputs.**
 - If the stereo to mono button is depressed, the L and R signals will be summed sent out the master left and master right xlr channels as mono signals.

- This section also controls balance. Both sides of a stereo system can be tested independently by turning this knob to either extreme. During the program, this knob should be pointed up at the 12 o'clock position to send the signal out both channels evenly.

33. ICM SE-4-ME 4CH Compact Audio Mixer. This audio mixer sends audio to the FM Transmitter. The input will either be direct from the main audio mixer, or it will be from the audio headend system. Once setup, levels to the FM Transmitter will be controlled by this mixer. Follow the AV designs provided. To use the mixer:



- (1) Connect the compact mixer to the main convention mixer, and to the FM transmitter.
- (2) Connect the USB drive with the FM announcements to the USB port (5).
- (3) Press the play/pause button (6) to play the announcements from the USB on a loop.
- (4) Lower the MP3 volume knob (4) to mute the announcements.
- (5) Raise the CH1 knob (1), the PHONES knob (3), and the MAIN OUT knob (2) to begin broadcasting the program audio.
- (6) Connect headphones to the PHONES jack (7) to confirm the audio is coming through the system.
- (7) Every hour, on the hour, lower the CH1 knob (1) and raise the MP3 knob (4).
 - Listen through the headphones, wait for the next announcement cycle to start and then play through.
- (8) Lower the MP3 knob (4), and raise the CH1 knob (1) to resume the program broadcast.

CHAPTER 4 VIDEO EQUIPMENT

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Media Players

1. There are different laptops that provided in the Australasia kits for the purpose of playing convention media.

- (1) HP i5 1235U Laptop.
- (2) Toshiba Tecra R850-10R i5 Laptop.

2. **Laptops for convention media.** The kit comes with two laptops. These are a Main and a Backup device for playing convention media. JW Library should be used to play all the convention media.

3. How to set up laptops for convention media.

- (1) Connect the laptop to power using the provided adaptor.
- (2) Connect the laptops to the AV system using the ATEN HDMI Converter and ProAV1 DI Boxes. See the section under Chapter 3, Audio Converters, 'Setting up the ATEN HDMI Converter and the ProAV1 DI Box with the media laptops.' For more information.

4. **If the output to the video switcher is black.** Often video switchers won't have an input scaler, meaning the input resolution needs to be correct for it to display properly. If there is no signal to the switcher, the troubleshooting steps are as follows:

- (1) Check that the resolution on the laptop under display settings matches the expected input resolution. The video switcher will have a label detailing the expected settings, but it will likely be 1080i50 or 720p50.
- (2) If the output settings are correct and there are still having issues, confirm there is an image of some sort being displayed on the second screen.
- (3) Confirm the converters are all powered on and working.
- (4) Confirm the cables and ports are all working (try using different ports and swapping cables).
- (5) Confirm the Active Signal resolution is correct.
 - Go to Display settings.
 - Go to Advanced display settings.
 - The desktop and active signal resolutions can be found under Display information.

Display information

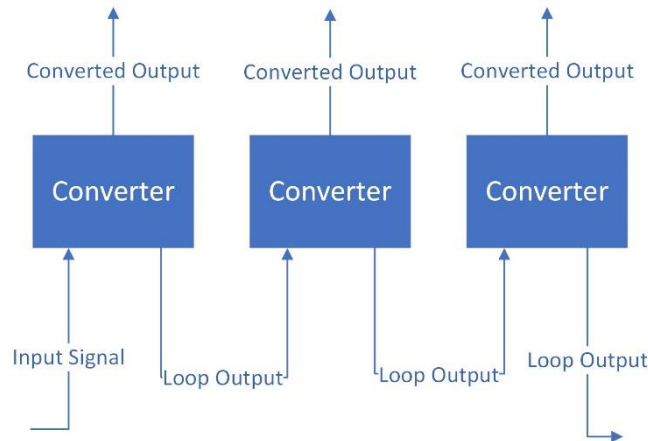
	DELL P2417H
	Display 1: Connected to Intel(R) HD Graphics 530
Desktop resolution	1920 × 1080
Active signal resolution	1920 × 1080
Refresh rate (Hz)	60.000 Hz

- Sometimes the active signal resolution will differ from the Desktop resolution that was set. If the values are different, this could be the cause of the problems.
- Try setting the resolution to a lower resolution, saving, then changing it back.
- If that doesn't work, try changing the refresh rate (from 50i to 25, or to 50).

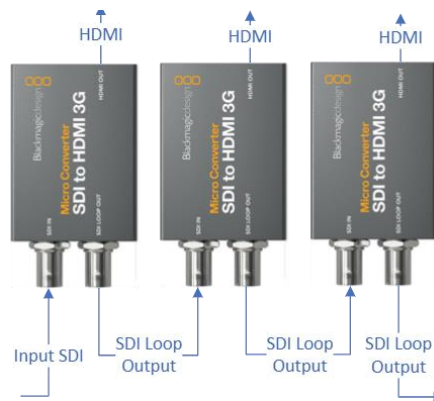
- If the input resolution can't be adjusted using either method, one of the Decimator MD-HX converters might be needed to handle video scaling to achieve the desired resolution, or contact the branch for support.

Video Converters

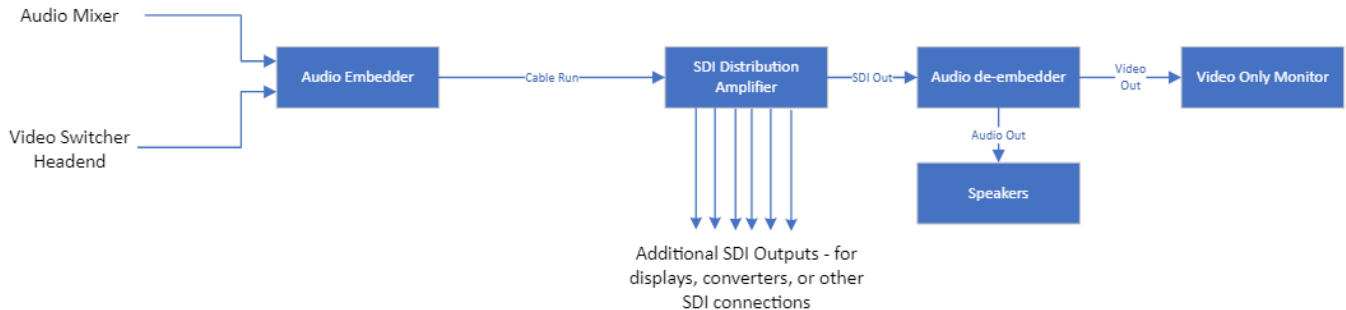
5. Video Converter Loop System. Most converters in the kit have a 'Loop' output. This allows multiple converters to be run off a single signal cable, by being connected together in series. If this approach is used, ensure the total cable length of each connection does not exceed a length of 100m, as this will result in signal degradation. Many video monitors also have loop out functionality.



- (1) An example would be using SDI to HDMI converters to make one SDI input signal run three HDMI displays.

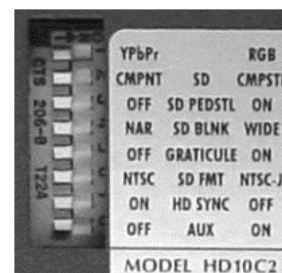
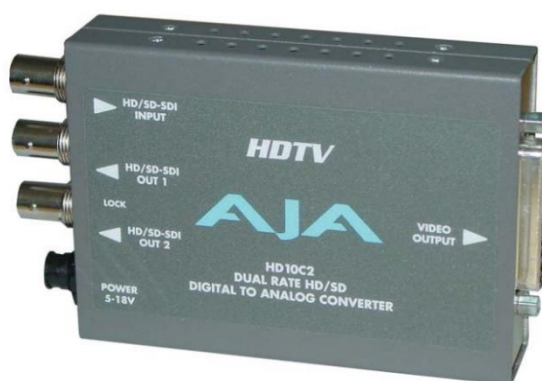


6. Using Audio Embedders and De-embedders. Often secondary video feeds will require an audio and a video signal to be transported to a distance away from the AV desk. To avoid the need to run both audio and video cables long distances, audio embedders and de-embedders are provided with the kit. This allows the program to be sent using a single cable. This embedded signal can be run through an SDI Distribution amplifier to create several embedded signals – and this can be done either at the AV Desk or after a cable run to a more appropriate location. Follow the AV design provided for the given venue.



- (1) Run the signal from the audio mixer and the video switcher headend into an audio embedder, this combines both the audio and video into an SDI signal.
- (2) Run the SDI cable from the audio embedder into an SDI Distribution amplifier.
- (3) Run the cable to where the monitors or speakers are needed.
- (4) Run the signal back through an audio de-embedder to separate the audio and video signals again.
- (5) Connect to the output devices.

7. AJA HD10C2 Video Converter. This is a Digital to Analog converter. The HD10C2 is a high-quality 10-bit HD converter that's "dual-rate" and works with both HD-SDI and SD SDI inputs. For HD-SDI inputs, the HD10C2 outputs full bandwidth HD component or "VGA" style RGBHV video. For SD-SDI inputs, component or composite SD outputs are supported. The HD10C2 also features 2 equalized HD/SD SDI outputs. The converter is configurable by a set of switches on the back of device, the description of each switch is provided below.



SWITCH	FUNCTION	DIP Set LEFT (default)	DIP Set RIGHT
1	Component Video Output Format	YPbPr: Selects YPbPr Component output (YPbPr)	RGB: Selects RGB Component output
2	SD Output Format:	COMPONENT: Selects Component output (SD Input)	COMPOSITE: Selects Composite output (SD Input)
3	SD Pedestal	OFF: Disable pedestal (SD Input)	ON: Enable pedestal (SD Input)
4	SD Blanking	NARROW: Select narrow blanking	WIDE: Selects wide blanking
5	Graticule: (HD Mode 4:3 Safe Area)	OFF: Disable graticule	ON: Enable graticule
6	SD Format	NTSC: (SD Input)	NTSCJ: Selects Japan's NTSCJ format (SD Input)
7	HD Sync	ON: Enable HD Sync	OFF: Disable HD Sync (remove sync on RGB)
8	Aux	Not currently used	

8. **Vcando HDMI to SDI converter.** This device takes an HDMI signal and converts it to an `SDI for use by the Video Switcher.



9. **Blackmagic SDI to HDMI Micro Converter.** This is used to display the program on a screen with only HDMI connections if needed. The device contains an SDI loop output to be connected to another converter or monitor if needed. If audio is needed at the monitor, make sure the converter is connected after the audio embedder system. The converters are supplied with power adaptors, but can be powered through USB from a computer or television.



10. **Blackmagic HDMI to SDI Micro Converter.** This is used to convert the HDMI signal from a laptop to an SDI signal for the Video Switcher Headend. The 'ATEN HDMI to 3G/HD/SD-SDI Converter' is the primary converter used for this purpose, with the Blackmagic devices serving as backup converters. This device does not automatically switch resolutions, and so there is a need to set the resolution appropriately on the laptop before running it through the converter. The Video switchers are labelled with the correct

resolution, but it will be either 720p50 or 1080p50. When using this device for convention media, use the audio jack on the laptops to connect into the audio system.

- (1) For information about the ATEN configuration, see Chapter 3 – Audio Converters – ATEN HDMI to 3G/HD/SDI Converter.



11. Decimator MD-HX HDMI/SDI Cross Converter. This device can convert from HDMI to SDI, or from SDI to HDMI. Generally, this would be used to convert HDMI into SDI at a given frame rate and resolution to be compatible with the video switcher.



- (1) **Conversion modes.** The cross converter can be used in the following modes of operation.
 - HDMI to SDI while simultaneously converting SDI to HDMI.
 - HDMI to HDMI while simultaneously converting SDI to SDI.
 - HDMI to SDI and HDMI.
 - SDI to SDI and HDMI.
- (2) The Decimator opens by default to the Input Status menu. This will show the status of the connected inputs and the configured output resolution and frame rate.

Input Status: (Has no SUB-MENUs)

Input Status	← Main Menu
S: 3GB 1080p59.94	← Current SDI input format
H: ED 480p59.94	← Current HDMI input format
D: HD 1080i59.94	← Current Scaling output format

The input shows the current SDI and HDMI input status as well as the DUC output format status.

- (3) Use the left and right buttons to navigate the menu.
- (4) To set the source for a given output (HDMI out or SDI Out) enter the control menu. Press enter and press right until the desired output setting is found. Then keep pressing enter until the desired input is found. Select HDMI or SDI for a straight conversion,

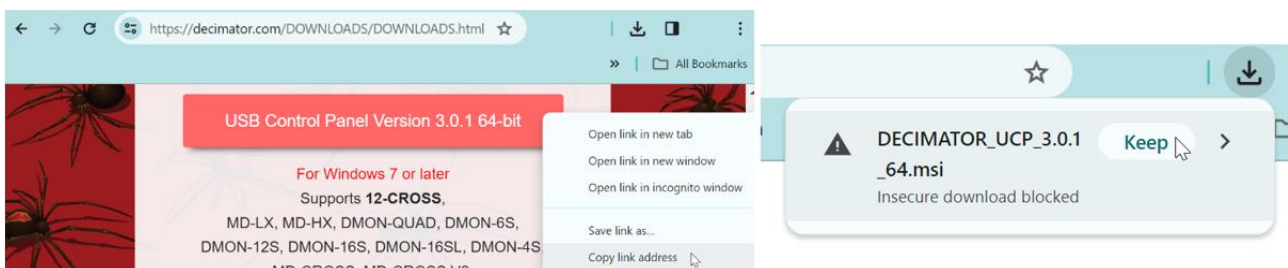
or select the scaler if a change is needed with the resolution. If a scaler is being used, enter the scaler menu and set the input and configuration for the scaler accordingly.

- (5) The input for the scaler is also set in control, under the sub menu DUC Source. Set it to be either SDI In or HDMI In.
- (6) To alter the output format for the scaler, press back twice if not in the main menu already. Then use arrows to navigate to the Scaling menu. Then press enter and navigate to the Output Format menu. Then keep pressing enter until the desired resolution settings are found.
- (7) With each change, an S will appear on the top right for a few seconds as the configuration saves. Endeavour not to cut power to the device as the configuration is saving.

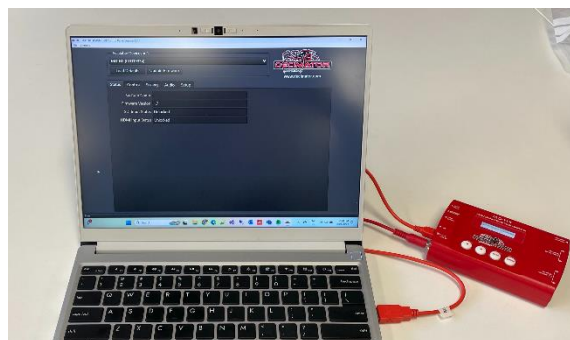


12. Configuring the Decimator MD-HX HDMI/SDI Cross Converter using a computer:

- (1) Configure the device using the buttons on the device itself, or configure the Decimator using a computer as this may be easier. This is particularly the case if several devices need to be configured the same way, or if there are advanced configuration requirements.
- (2) Download the correct Decimator USB Control Panel software for the given operating system.
 - Note: The downloads page for the Decimator software uses HTTP and not HTTPS so the browser might try to prevent the software from downloading. If that happens, manually copy and paste the download link into a new tab. Then press 'Keep' on the download prompt that pops up.



- (3) Connect the Decimator device to power and to the computer.



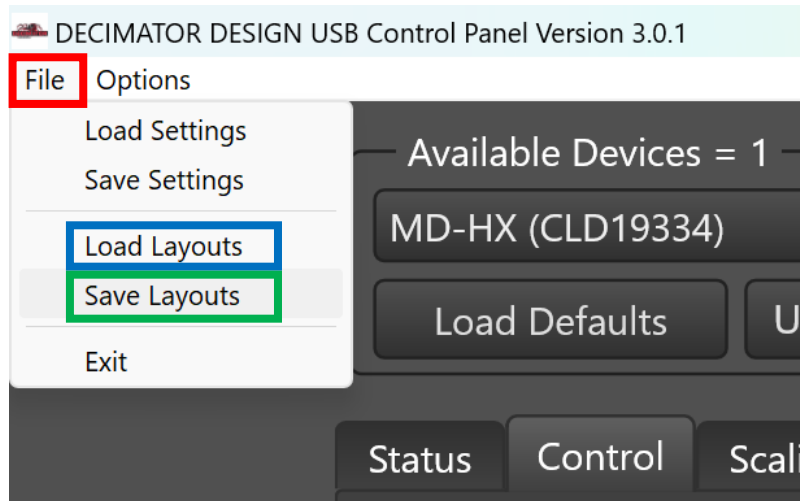
- (4) Open up the control panel software (it shows up as UCP in the search bar).
- (5) Use the control panel to configure the device.

- (6) The Decimator shows the same S on the led display when it is saving changes. Don't close the software or disconnect the device when that symbol is displaying.



13. Applying the configured Decimator settings to other devices using the control panel.

- (1) Use the UCP control panel software to configure the first Decimator device.



- (2) Press **File** -> **Save Layouts** to create a save file (this may take a few minutes).
- (3) Once the file has finished saving, connect the next Decimator device.
- (4) Press **File** -> **Load Layouts**.

14. **Decimator MD-LX HDMI/SDI Bi-Directional Converter.** This device can convert from HDMI to SDI or from SDI to HDMI. The conversion modes are automatic depending on what inputs are connected.

- (1) If just SDI input is connected, it will convert and send that signal to both the HDMI and SDI outs.
- (2) If just HDMI input is connected, it will convert and send that signal to both the HDMI and SDI outs.
- (3) If both HDMI input and SDI input are connected, the SDI signal will feed the HDMI out and the HDMI signal will feed the SDI out, operating in bi directional conversion mode.



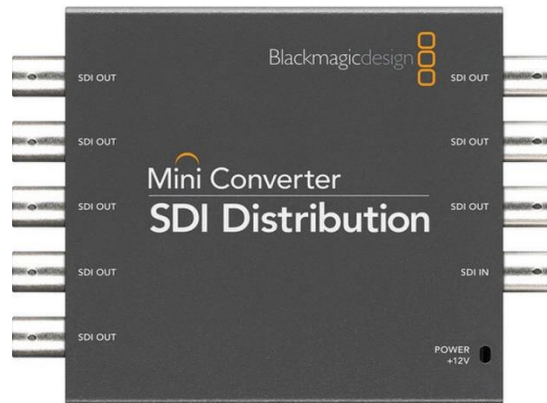
Distribution Amplifiers

15. Distribution amplifiers are used to take a single SDI signal and send it to multiple output channels. Generally, these are connected after the audio embedder, so that all the outputs will have audio included without needing to run additional cabling (whether audio is needed at every output or not).

16. **AJA 3GDA Distribution Amplifier.** This device is used to take a single SDI signal and distribute it to six SDI output signals.



17. **Blackmagic 1 to 8 SDI Distribution Mini Converter.** This device is used to take a single SDI signal and distribute it to eight SDI output signals.



Tripods

18. **Miller Cat1511 2 Stage Tripod.** This carbon fibre tripod is provided to support the manned Panasonic HPX-250 Video Cameras.



Camera Systems

19. **Panasonic HPX-250:** The Panasonic HPX-250 is a fixed-lens, manned camera.

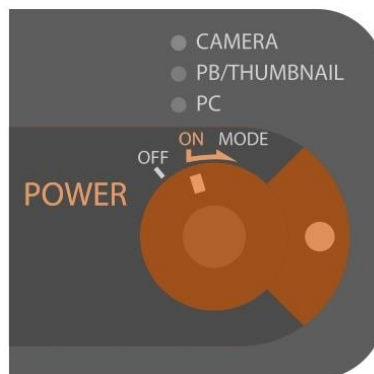


20. How to Set Up the Panasonic HPX-250:

- (1) Mount the camera on the tripod by sliding the attached plate on the bottom of the camera into the matching grooves on top of the tripod head until it clicks into place. Tighten the locking knob to secure the camera.
- (2) Connect the camera power brick to the camera by inserting the connector into the battery compartment and sliding it down.
- (3) Connect an SDI cable from the SDI OUT port to the video switcher.

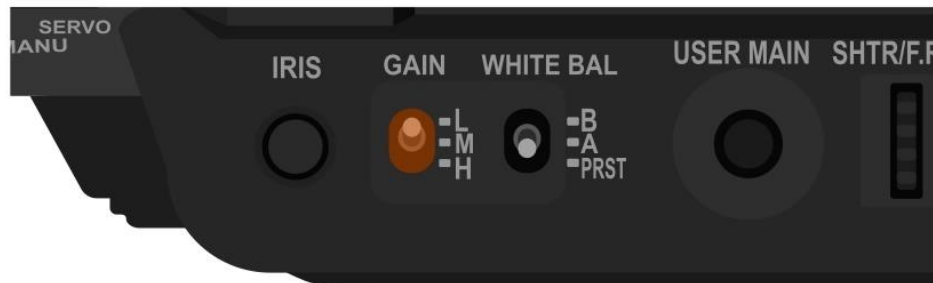


- (4) Remove the camera's lens cap.
- (5) Turn on the camera.



21. **Gain:** In some venues, there is not enough light on the stage to get a bright enough picture, even with the iris fully open. In this scenario, raise the gain on the camera until it is bright enough to provide a clear picture. However, because gain adds noise to the picture, it is recommended to keep it as low as possible.

- (1) To adjust the gain, toggle the GAIN switch on the side of the camera, shown below.
- (2) When Adjusting gain, always try the lowest levels first and go up as needed. Only if even at max IRIS the image is still dark, should gain be increased.



22. **White Balance:** To manually set the white balance on the camera, flip the toggle switch on the side of the camera under WHITE BAL to A.



- (1) Hold up an a3 sheet of white card or paper on stage to fill most of the shot.
- (2) Press the AWB button located under the lens on the front of the camera.



- (3) The screen will display AWB A ACTIVE. Once the white balance is set, it will show AWB A OK with a numerical value.

23. If absolutely necessary, auto tracking white balance (ATW) can be used at outdoor venues, as the colour temperature may shift throughout the day. However, manual white balance is recommended. The ATW function can be assigned to the white balance switch 'B' by setting the menu option 'ATW' in the 'SW MODE' screen to 'B ch'. With this configuration, toggling the WHITE BAL switch to B will enable auto white tracking white balance.

24. **Zoom.** The Zoom operation should be done through the zoom switch (3). To enable this functionality, the Zoom Mode (1) should be in SERVO mode. A soft press will zoom slowly, and a hard press will zoom quickly. The HANDLE ZOOM (2) switch will also adjust the speed of the zoom.



25. **Focus.**



- (1) The FOCUS Switch (2) toggles between automatic and manual focus mode. Setting the focus switch to infinity will make the focus go to max distance and then toggle back into manual mode.
- (2) When in manual mode, PUSH AUTO will make the camera quickly auto focus.
- (3) When in manual mode, toggling FOCUS ASSIST will enable a hybrid manual/auto mode. In this mode of operation, coarse focus will be adjusted using the manual ring, and fine adjustments will be made automatically.

26. **Iris.** This adjusts the brightness of the image. If the image is not bright enough with a fully opened iris, see the Gain section for information on raising the brightness further.



- (1) Pressing the IRIS button (1) in will toggle between auto and manual Iris mode.
- (2) The IRIS ring (2) can be rotated to adjust iris when in manual mode.

27. **Normal Operation.** For normal operation, check that the camera switches are all set as follows:

- (1) ND FILTER set to OFF.
- (2) FOCUS set to A.
- (3) ZOOM set to SERVO.
- (4) GAIN set to L.
- (5) WHITE BAL set to A.
- (6) Switch set to MANU.



Camera Tally Systems

28. **AV Matrix Wireless Tally system.** The two main components of this system are:



- (1) AV Matrix Tally Light.
- (2) AV Matrix Tally Box.

29. Setting up the AV Matrix:

- (1) Screw the lights into the mounting points on each of the cameras. Ensure the correct numbered light goes onto the correct camera (light one to camera one, etc.).



- (2) Connect each tally light to power via USB c.
- (3) Connect the yellow Tally Box up.
 - Connect it to power via USB c.
 - Connect it to the video switcher using the WTS 02 001 cable.



- Connect it to the antennae.
- (4) DIP switch configuration.
 - The tally box and tally lamp DIP configurations should not be changed without direction or approval from the branch.
 - The tally box should come with every switch but number 3 in the on position, as shown below.



- The DIP switches for the box operate according to the following table.

WIRELESS TALLY SYSTEM					
DIP (1=ON, 0=OFF)					
INPUT	SW1	SW2	RS-485	SW3	GROUP
GPI0	1	1	Input	1	Group 1
USB-C	1	0	Output	0	Group 2
RS-485	0	1	MODE	SW4	USB-C
RS-232	0	0	DIP	1	Working
			PC	0	Config
					SW6

- The DIP switches for lamp are used to configure their operation mode, and which camera number they relate to. These are also configured by the branch.

- The two DIP switch sets for the lamp are found inside the battery compartment.



- These two sets of switches are configured differently for each unit, and the behaviour is defined by the following table.

DIP 1 (1=ON, 0=OFF)					DIP 2 (1=ON, 0=OFF)	
LAMP	SW1	SW2	SW3	SW4	INPUT	SW1
NO. 1	0	0	0	0	Wireless	1
NO. 2	1	0	0	0	RS-485	0
NO. 3	0	1	0	0	MODE	SW2
NO. 4	1	1	0	0	DIP	1
NO. 5	0	0	1	0	PC	0
NO. 6	1	0	1	0	GROUP	SW3
NO. 7	0	1	1	0	Group 1	1
NO. 8	1	1	1	0	Group 2	0
NO. 9	0	0	0	1		
NO. 10	1	0	0	1		
NO. 11	0	1	0	1		
NO. 12	1	1	0	1		

Video Monitors

30. **Samsung S20D300HY Computer monitor.** This is a 20 inch monitor. Connect it to the Video Switcher to provide a multi-view display for the Video Switcher Operator and Technical director.

31. **Baas & Kuhn MUDV43 Video Monitor.** This is a 43" monitor. For Conventions use the SDI input labelled SDI IN.



32. Setting up and using the MUDV43 monitors.

- (1) If the monitors are going to be free standing, install the monitor feet. Use the silver Monitor Stand screws to secure the feet properly.



- (2) If the monitors are going to be wall mounted, use the VESA mount on the rear of the monitors. The convention's AV Department will provide a bracket to be used. The screws are in a bag labelled 'Wall Mount Screws'.



- (3) The monitors have an SDI Loop output, which means one input source can be used to run multiple monitors by daisy chaining them.

- Connect an SDI cable from the input to the first monitor in the chain.
- Connect an SDI cable from the SDI Out of the first monitor, into the SDI In of the next monitor.
- Repeat until all monitors are connected.



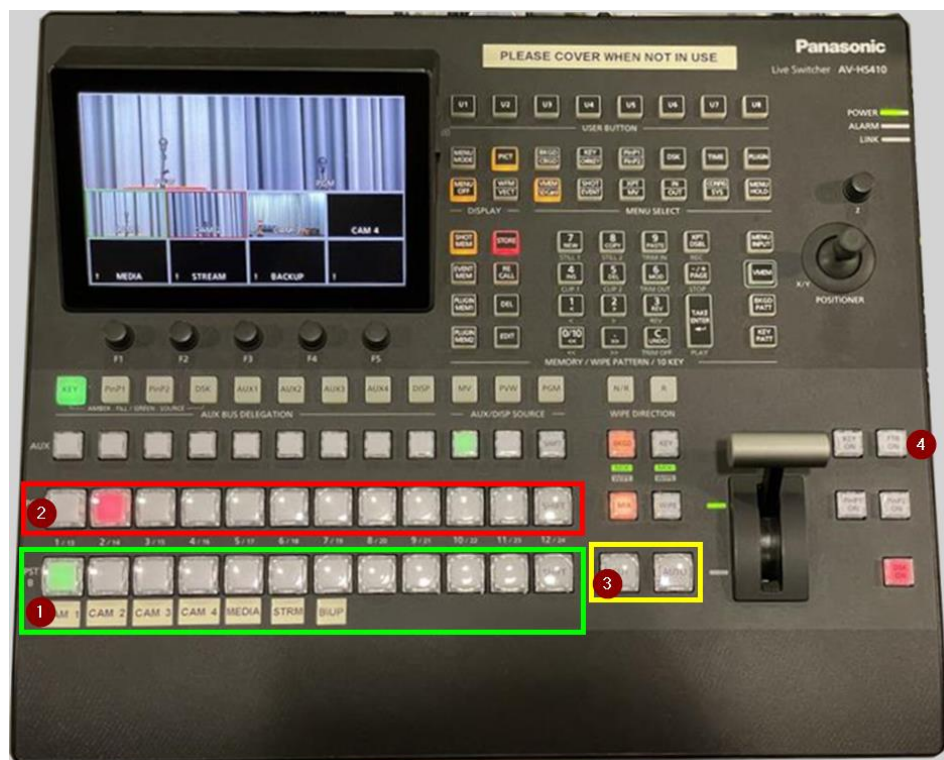
Video Switcher Headend

33. The standard video switcher headend contains:

- (1) 1x Furman Power Conditioner.
- (2) 1x Panasonic AV-HS410 Video Mixer.

Video Switchers

34. **Panasonic AV-HS410 Video Switcher.** The controls highlighted in the below picture are the ones that will be used for conventions.

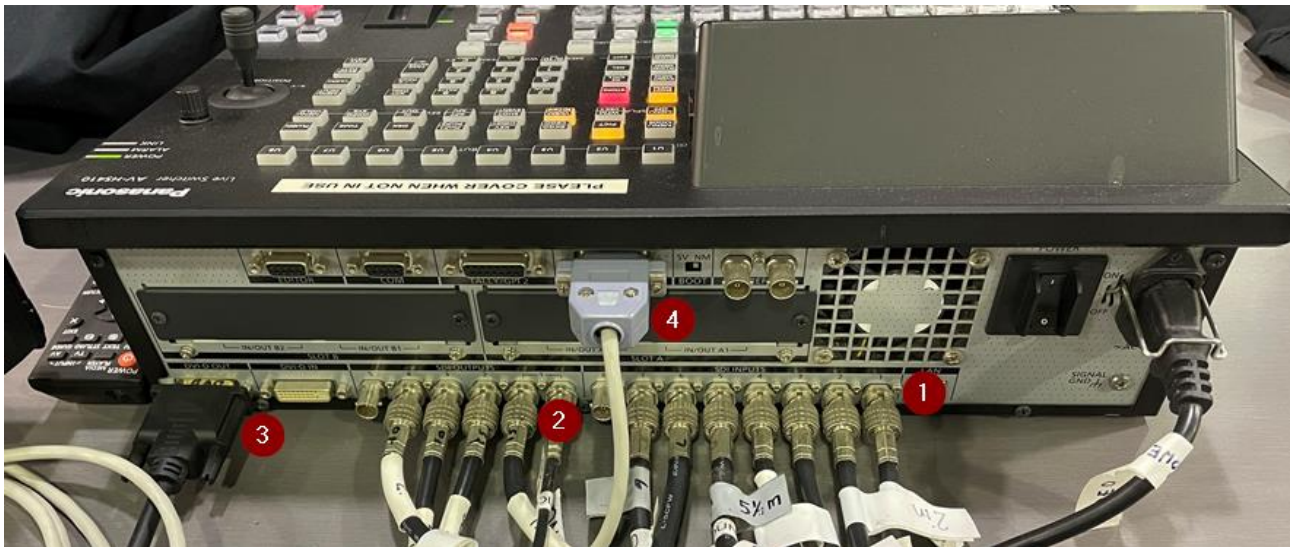


- (1) **The Preview section.** Pressing these buttons will make the input show in preview (top left). Pressing cut or auto will make the previewed signal go live.
- (2) **The Live section.** Pressing these buttons will make the input go directly to live, skipping preview. This should only be used in an emergency (e.g. falling to media backup).

- (3) **The Cut/Auto section.** These two buttons swap the live and preview signals, making the previewed input go live. Cut makes the input instantly replace the live signal. Auto makes the one feed fade into the other.
- (4) **FTB ON.** This button makes the screen fade to or from black, which can be useful in emergencies, or in prayers when a wide-angle shot is not available. An alternative to using this button is to auto into one of the unused inputs.

35. Setting up the Panasonic Video Switcher.

- (1) Connecting the Video Switcher.



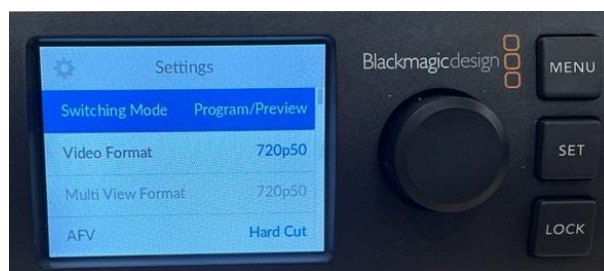
- **SDI Inputs (1).** Connect the SDI inputs here. Ensure the order lines up with the stickers on the switcher, and the names shown in the preview windows.
 - **SDI Outputs (2).** There are 6 SDI Output feeds available for use. Use as many as needed, following the AV designs provided.
 - **DVID OUT (3).** This DVID out port is used to power a multi-view monitor for the video switcher, PTZ operator and technical director to all work off. A DVID to HDMI adaptor might be needed depending on the available monitors for the AV Desk.
 - **TALLY/GPI 1 (4).** This port is used to connect into the tally light box.
- (2) The inputs should each be 720p50. If any inputs have a '!' icon, this indicates the signal either isn't connected or that the input resolution is incorrect.



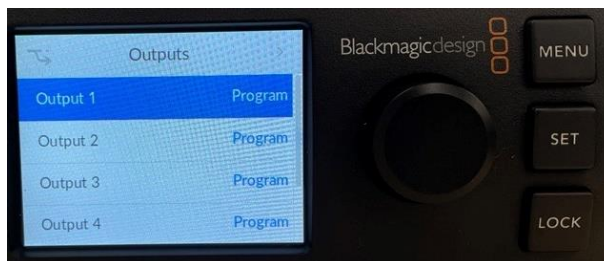
36. **Atem video switcher constellation 1 me hd.** This switcher's main function in conventions is as an audio embedder. The device can also function as a backup video switcher to replace the Panasonic system when necessary.



- (1) There are a few configurable settings to the switcher. These should not be altered without direction or approval from the branch.
- (2) To view or change the output video format, press the menu button to enter the menu. Then use the scroll wheel to navigate to the Video Format setting. Press SET to enter the setting, scroll to adjust the value, and set again to confirm the change. This was set to 720p50 by the branch.



- (3) The outputs on this device are configurable through the Menu. Each can be set to display the program output, a multi-view view, or a number of other options. From the branch these are all set to display the Program output. To adjust these, scroll down to the 'Outputs' option in the main settings menu, and press SET to enter this sub-menu. From here each of the 6 custom outputs can be configured.



- (4) When not in a menu, the knob will control the audio out levels.

37. Atem Video Switcher as an audio embedder and SDI distribution amplifier. This mode of operation will involve only using the embedding functionality of the switch, and only having video input coming through a single input channel.

- (1) Connect input 1 to one of the program outs of the Panasonic video switcher.
- (2) Connect the audio from either the audio headend, or direct from the main audio mixer (follow the AV designs provided).
- (3) Make sure input 1 is the live input (red) and is showing in the preview window.
- (4) SDI Outputs 1-6 should all now be sending out the embedded program signal, and can all be used as needed.

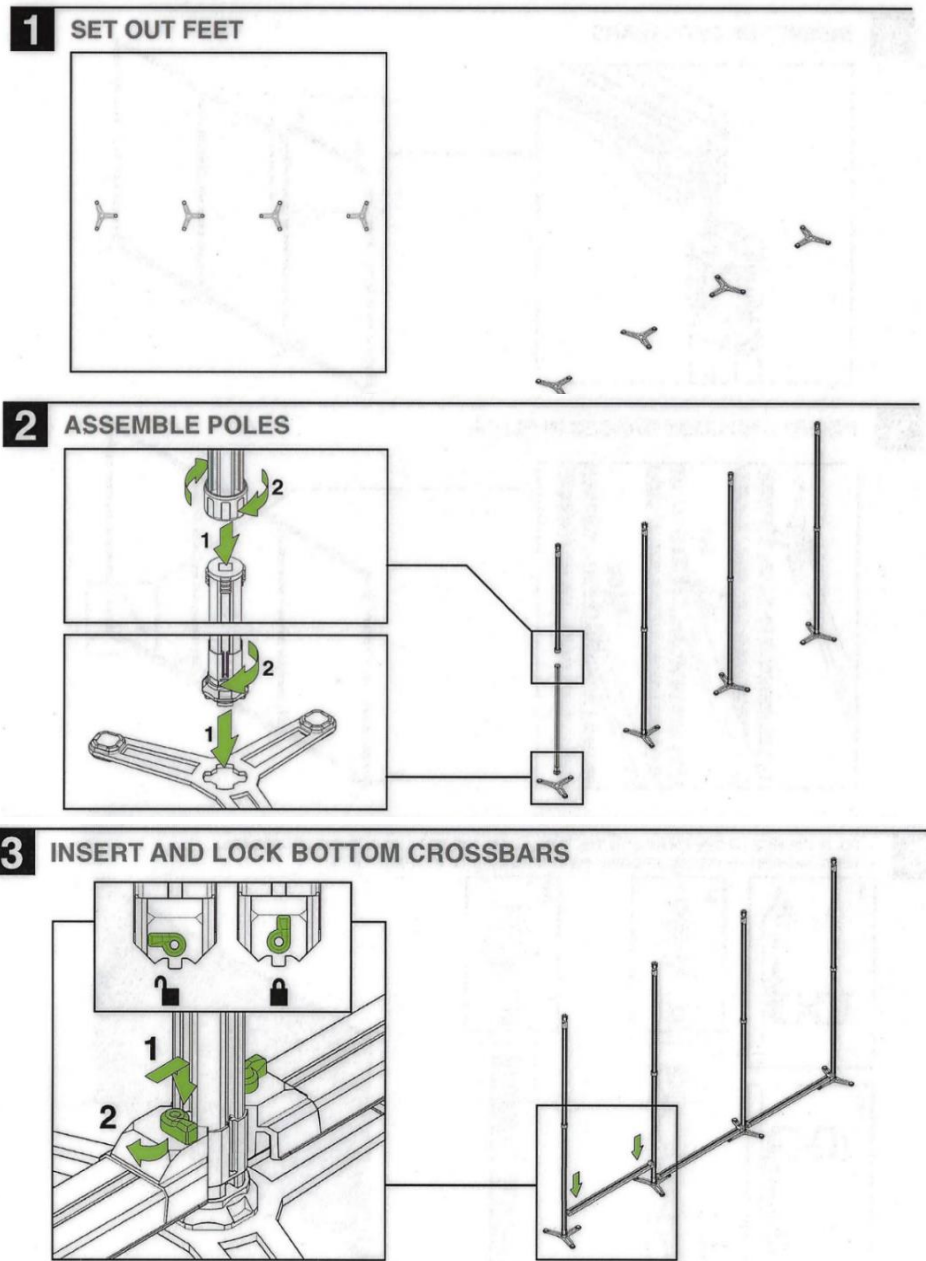
38. Atem Video Switcher as the main video switcher.

- (1) Connect the Multiview port to an available monitor.
 - The Panasonic monitor in the kit only has DVI input. To connect this to the Multiview will require conversion.
 - Conversion can be done by using one of the provided SDI to HDMI converters, and then an HDMI to DVI adaptor to connect the two devices.
- (2) Connect the inputs to the video switcher.
- (3) Use the switch as normal.
 - Pressing an input button will put it in preview mode (green).
 - Pressing CUT or AUTO will make the preview input go live (red).
 - Pressing FTB will make the input fade to or from black. It can be safer to instead use AUTO to fade into an unused input like 10, but the FTB button is always available for emergencies.

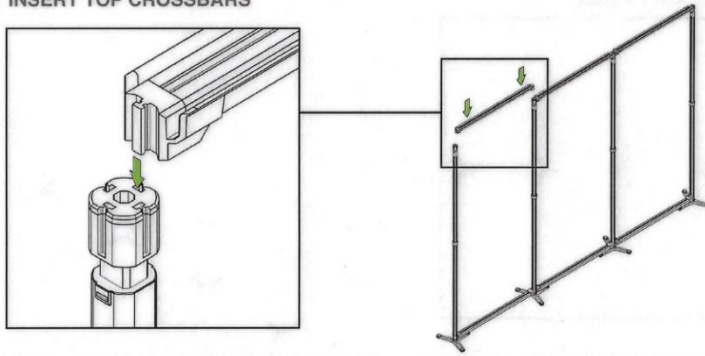
Backdrops

39. **Convention Backdrop Set.** This is the equipment that provides the blue background for the speakers/stage. **Do not lay the curtain on the floor/ground without protecting it from dirt and moisture nor wash the curtain.** Use a brush to remove dust or dirt and use only a steam iron to flatten the curtain when needed.

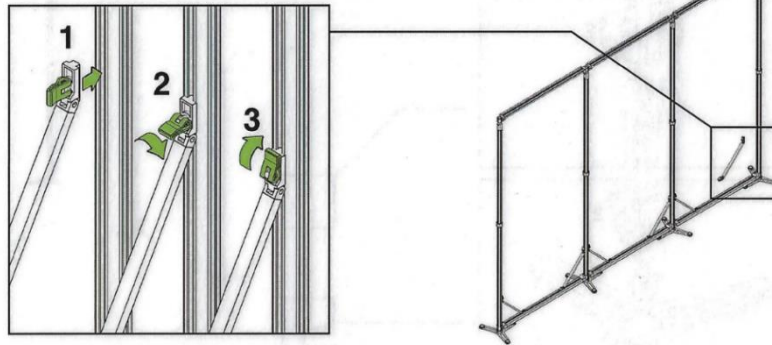
40. Setting up the Convention Backdrop.



4 INSERT TOP CROSSBARS

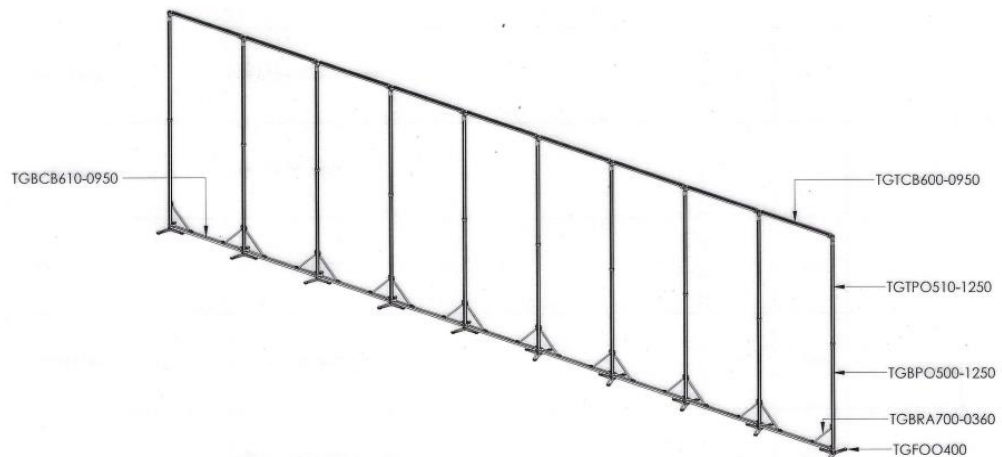
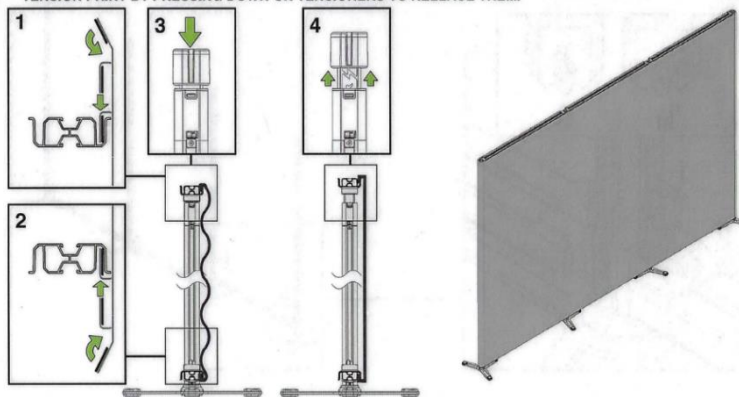


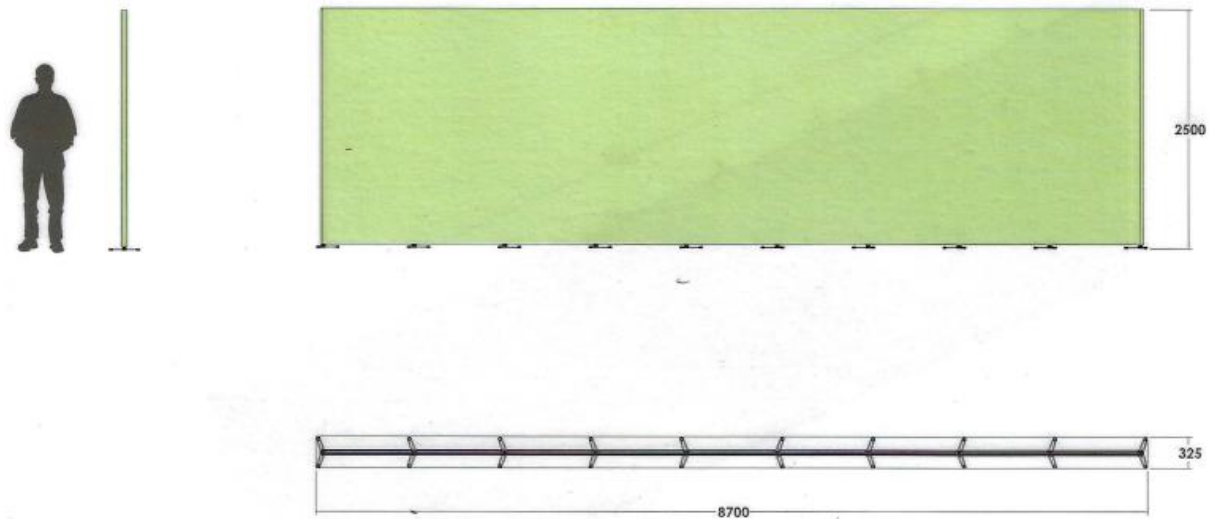
5 INSERT AND LOCK BRACES IN PLACE



6

- FOLD THE HEM OF THE PRINT INTO THE TRACKS ON THE TOP AND BOTTOM CROSSBARS.
- TENSION PRINT BY PRESSING DOWN ON TENSIONERS TO RELEASE THEM.





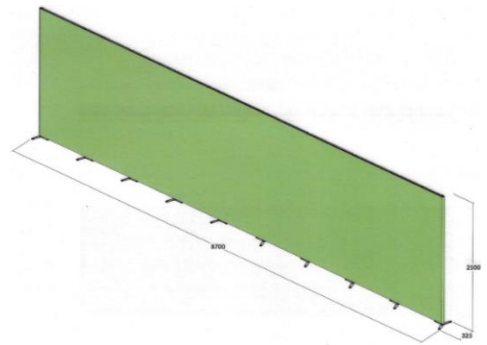
41. Set up the side backdrops on both sides.



42. Sandbags can be purchased locally if required, and used at the base of the backdrop.

43. Backdrop Reference Pictures:





44. Packing up the backdrop.



45. Backdrop Fire Rating.



Australian Wool Testing Authority Ltd - trading as AWTA Product Testing
A.B.N 43 006 014 106
1st Floor, 191 Racecourse Road, Flemington, Victoria 3031
P.O Box 240, North Melbourne, Victoria 3051
Phone (03) 9371 2400

TEST REPORT

Client : Cortex Pty Ltd
1B, 26-28 Phillips Road
Kogarah NSW 2217

Test Number : 22-001944
Issue Date : 3/06/2022
Print Date : 3/06/2022
Order Number : NV30522_2

AS 1530.2-1993

Methods for Fire Tests on Building Materials, Components and Structures.

Part 2: Test for Flammability of Materials

Date Tested	03-06-2022		
Flammability Index	Length	Width	
	0	0	
Spread Factor	1	1	
Heat Factor			
Maximum height (d)			
Mean	1.4	1.7	
Coefficient of Variation	14.4	15.5	%
Heat (a)			
Mean	1.5	1.5	°C min
Coefficient of Variation	0.0	0.0	%
Number of Specimens Tested	6	6	
Observation	Visible smoke, melting, dripping.		

These test results relate only to the behaviour of the test specimens of the material under the particular conditions of the test, and they are not intended to be the sole criterion for assessing the potential fire hazard of the material in use.



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Test Number : 22-001944
Issue Date : 3/06/2022
Print Date : 3/06/2022
Order Number : NV30522_2

Sample Description Clients Ref : "S/D32"
Warp knit Banner/Frontlit
Colour : White
End Use : Exhibition/Frontlit
Nominal Composition : 100% Polyester
Nominal Mass per Unit Area/Density : 230g/m2



Fibre Rack

46. The fibre racks are used to transmit the SDI video signal and Cat5 for PTZ camera control across large distances with the 300-meter fibre cable reel. This is used in setups when the AV desk is located more than 90m away from the cameras.



47. To connect the fibre cable to the Rack, remove the cap on the fibre plug, unscrew the two sided thread on the cap of the fibre cable and line up the pin to plug the cable in. Screw the blue cover to secure the cable to the rack.



48. The fibre rack will be setup like the image below at the camera end.

- (1) Plug Cam 1 into “DATA 1” and “IN 1”
- (2) Plug Cam 2 into “DATA 2” and “IN 2”
- (3) Plug Cam 3 into “DATA 3” and “IN 3”



49. The fibre rack will be setup like the image below at the controller end.

- (1) Plug the PTZ controller into “DATA 5” with a Cat5e cable
- (2) Connect Cam 1 to the video switcher by connecting a SDI cable from “OUT 1” on the fibre rack to “INPUT 1” on the Video Switcher
- (3) Connect Cam 2 to the video switcher by connecting a SDI cable from “OUT 2” on the fibre rack to “INPUT 2” on the Video Switcher
- (4) Connect Cam 3 to the video switcher by connecting a SDI cable from “OUT 3” on the fibre rack to “INPUT 3” on the Video Switcher



CHAPTER 5 ELECTRICAL DISTRIBUTION EQUIPMENT

Power Distribution Unit	1-5
Jaycar MS4023 Power Distributor	1
DesignNest Power Cube	2
Site Power 415V/32A Power Distributor	3
Power Conditioners	6
Furman PL-PLUS C E Power Conditioner	6
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Power Boards	7
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Ansmann 'Energy Check LCD' Battery Tester	9
Cable Tester	10

Power Distribution Unit

1. **Jaycar MS4023 Power Distributor.** This vertical power distributor features 8 surge protected sockets, power overload protection and two built in USB ports. The max power draw is 2400W.



2. **DesignNest Power cube.** This device is used to power the cameras. The power adaptor for the cameras can connect to the outlets of the cube, and the Tally Light lamps can be powered through the USB outs of the power cube.



3. **Site Power 415V/32A Power Distributor.** The connections to the main Audio/Video equipment should be through this power Distribution Equipment.

- (1) The distributor is designed to protect against power surges and dips. By connecting to this system instead of directly to mains, the risk of damaging any equipment is reduced.
- (2) The Distributor has its own breaker switches as well to make resolving issues easier.



4. When everything is connected appropriately, power up the equipment to confirm normal operation. Once the equipment is powered on at the start of each day, it is left running throughout the sessions and only turned off at the end of the day after the session has finished.

5. If any equipment does not power up as expected, first check all of the connections and then ensure that no circuit breakers have tripped on the sound distribution board. If none of the equipment is operating, supply from the main switchboard to the sound distribution board should be checked to ensure it is powered up and that none of the venue's circuit breakers have been tripped.

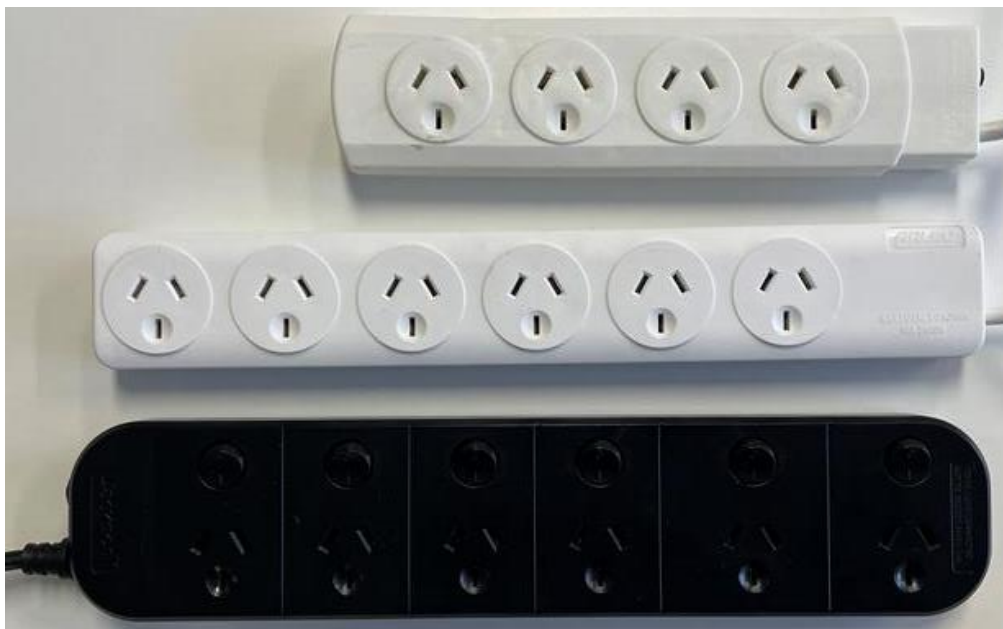
Power Conditioners

6. **Furman PL-PLUS C E Power Conditioner.** This powers the audio and video headend systems. The device stabilizes and cleans incoming electricity to protect the AV equipment, guarding against surges and noise.



Power Boards and Cables

7. **Power Boards.** The Kit contains several different power boards for use. Avoid chaining power boards together as that risks overloading the boards and creates a fire risk. Avoid placing power boards in high traffic zones as they can be a trip hazard. Each power board in the kit can safely handle up to 10A 2400W. If the power board suddenly stops working, it could be that it was overloaded and the board's circuit breaker tripped. If this happens look for the black button at the base of the board, and try pressing it in to reset the breaker.



8. **Power Extension Cables:** The kit comes with extension cables of different lengths. To avoid safety risks, do not connect multiple cables to extend their length. Instead, opt for the longer cables provided, which are designed for safe current transmission over extended distances. If extension cables must be run across areas with foot traffic, use the included cable protectors to prevent trip hazards.

Testing Equipment

9. **Ansmann 'Energy Check LCD' Battery Tester.** This is used to verify any batteries are still functional when setting up the convention equipment. Place the battery into the appropriate slot, and apply the pin to the other end of the battery when relevant. Within a few seconds, the tester will get a readout on the capacity of the battery.



Cable Tester. The cable tester is used to test various cables. Each conductor of the cable can be tested one at a time to ensure continuity. The tester does this by sending a signal down one channel at a time, and allowing the output to be read.

- (1) To use, connect the cable to be tested to both appropriate connections on the tester.
- (2) Cycle the knob to test each conductor (1-3 for a three pin cable, 1-5 for a five pin, etc.).
- (3) If any conductor does not activate the orange light, it means there is a broken conductor or connector. If any conductor activates two orange lights, one conductor is shorting into another. In either case, mark the cable as defective and do not use it.



Appendix A - Simultaneous Interpretation

INTERPRETATION SETUP

1. To accommodate simultaneous interpretation at select events, venues will receive an interpretation kit that includes (1) Xenyx mixer, (1) Rane headphone amplifier, (1) RDL distribution amplifier, (3) Audio-Technica interpreter headsets, (1) Sony headset, various power supplies and converters.
2. The simultaneous interpretation room should be enclosed to limit noise from the convention. If possible, it should be located close to the video crew, since several audio/video connections between the two areas are necessary.
3. **Behringer Xenyx QX1222 USB Mixer:** The Xenyx mixer is used to mix the interpreters voices and send this interpreted mix to listeners.



4. How to Set Up the Behringer Xenyx QX1222 USB Mixer:

- (1) Refer to the drawing detail "Simultaneous Interpretation" for information on connecting the interpretation mixer to the audio system.

- (2) The mixer operator's Sony headset should be plugged into the PHONS/CTRL jack on the mixer.



- (3) The headphone volume is adjusted using the PHONES knob, shown below.



5. **Operating the Behringer Xenyx QX1222 USB Mixer:** Raise and lower the slider for the corresponding interpreter or media input in order to send the audio to those listening to the translated program.

6. **Rane HC6S Headphone Amplifier:** This provides audio to the interpreter headsets. It allows each interpreter to control the volume at which they hear the audio from the main program.



7. Setting up the Rane Headphone Amplifier:

- (1) The front panel switch must be set at MONO.



8. Operating the Rane Headphone Amplifier:

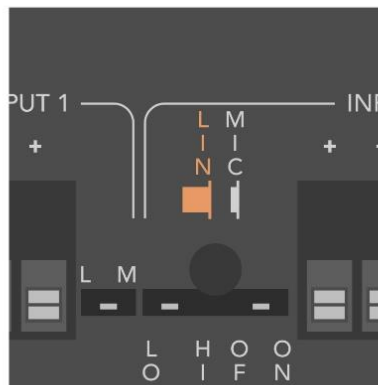
- (1) Each interpreter can raise and lower the volume to their headset by adjusting the corresponding knob.

9. RDL RU-MLD4 Distribution Amplifier: The RDL RU-MLD4 distribution amplifier supplies line-level audio to the headphone amplifier and the Mackie mixer.



10. Setting Up the RDL RU-MLD4:

- (1) Ensure the button on the back of the RDL RU-MLD4 is set to LIN. The button will be up.



INTERPRETATION OPERATION

11. The interpreters and mixer operator will sit at a table. Each interpreter will be provided with a broadcast headset. These headsets will be connected to a headphone amplifier that allows each interpreter to individually adjust the volume at which he or she hears the program. Each headset also has an attached microphone. Place a video monitor on the table so that both the interpreters and the mixer operator can see the convention program as it is displayed to the audience.

12. This system supports three simultaneous interpreters:

- (2) The primary interpreter translates for the chairman and for those who give talks.
- (3) The male interpreter translates for other male program participants, such as those who deliver interviews, soliloquies, or demonstrations.
- (4) The female interpreter translates for all female program participants.

13. The mixer operator will not hear the primary convention language. His headphones will be connected only to the mixer. This will allow him to carefully monitor and adjust the level and equalization of the interpreters. When music is played, he will raise the mixer input that receives audio from the primary convention sound system. When videos are played, he will raise the mixer input that receives the alternate language feed from the primary media player.

14. The mixer operator should keep levels balanced throughout the program. Interpreter microphones should be turned off when the interpreter is not speaking.